

W05 · M05 · 15 MINIT / MINUTES · Hari 2, 10:10 hingga 10:25

Hands-On SQA-AI Assistant Workshop

AI-generated UI and API tests that find a real bug

Ujian UI dan API dijana AI yang menemui pepijat sebenar

Ask the AI for a Playwright test for the quantity boundary, save and run it against ShopFast, read the failure, then generate API tests from the OpenAPI contract.

opencode 1.18 · model Big Pickle (OpenCode Zen, free, no API key) · synthetic ShopFast data only

Bengkel SQA KRISAv2 dan PPrISA 2.0 · 14 hingga 15 September 2026 · Makmal Komputer Aras 11, Menara MyIPO

[Web guide \(copy prompts\)](#)

[Module hub](#)

[Example outputs \(GitHub\)](#)

Before you start / Sebelum mula

YOU WILL / ANDA AKAN

1. Run `/gen-ui-test` for TC-CHK-004 (quantity 10)
2. Save the code block as `tests\ui\m05-senario.spec.js`
3. Run it with `npx playwright test` and read the result
4. Run `/gen-api-tests` for the same requirements

1. Jalankan `/gen-ui-test` untuk TC-CHK-004 (kuantiti 10)
2. Simpan blok kod sebagai `tests\ui\m05-senario.spec.js`
3. Jalankan dengan `npx playwright test` dan baca hasilnya
4. Jalankan `/gen-api-tests` untuk keperluan yang sama

CHECK FIRST / SEMAK DAHULU

- PowerShell is in `C:\Users\<you>\sqa-krisa-bengkel\lab`
- `opencode` starts and shows Big Pickle, OpenCode Zen (W00)
- In `shopfast\`: `npm install` and `npx playwright install chromium` are done

Data rules / Peraturan data

- Synthetic ShopFast data only. Never paste real agency documents or personal data.
- Free models run outside Malaysia and may learn from prompts.
- AI output is a draft. A person checks and signs off.

OUTPUT

`outputs/m05-ui-test.md`, `shopfast/tests/ui/m05-senario.spec.js`, `outputs/m05-api-tests.md`

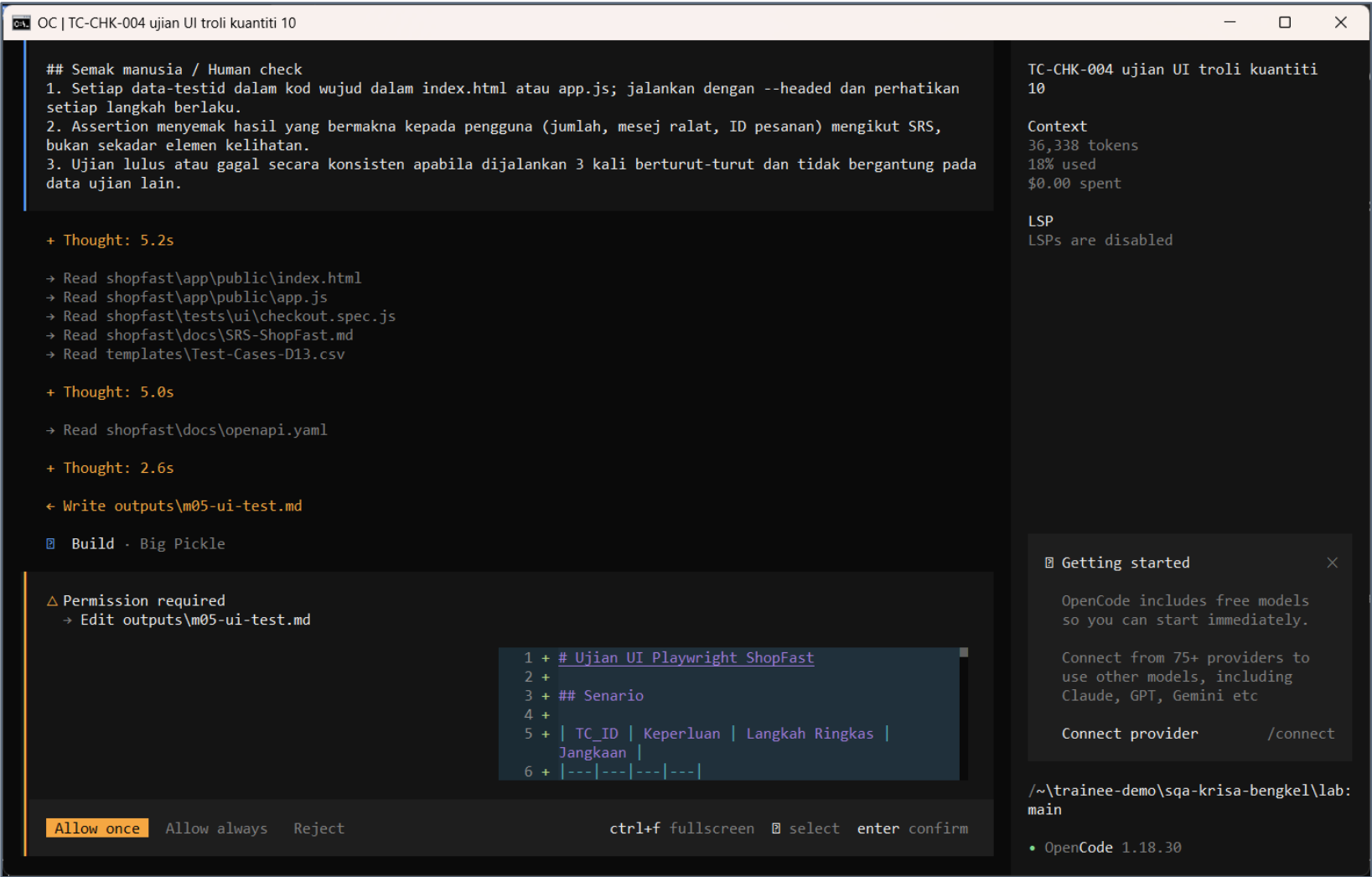
W05 · STEP 1 OF 5 · LANGKAH 1

Run /gen-ui-test for the quantity 10 boundary case. The AI reads the documents first and may pause at Preparing write. When the Permission required dialog appears, read the preview and choose Allow once.

Jalankan /gen-ui-test untuk kes sempadan kuantiti 10. AI membaca dokumen dahulu dan mungkin berhenti seketika pada Preparing write. Apabila dialog Permission required muncul, baca pratonton dan pilih Allow once.

TYPE IN OPENCODE (BUILD) Salin / Copy

```
/gen-ui-test TC-CHK-004 tambah P002 kuantiti 10 ke troli dan semak troli menerima kuantiti 10 (REQ-CHK-01)
```



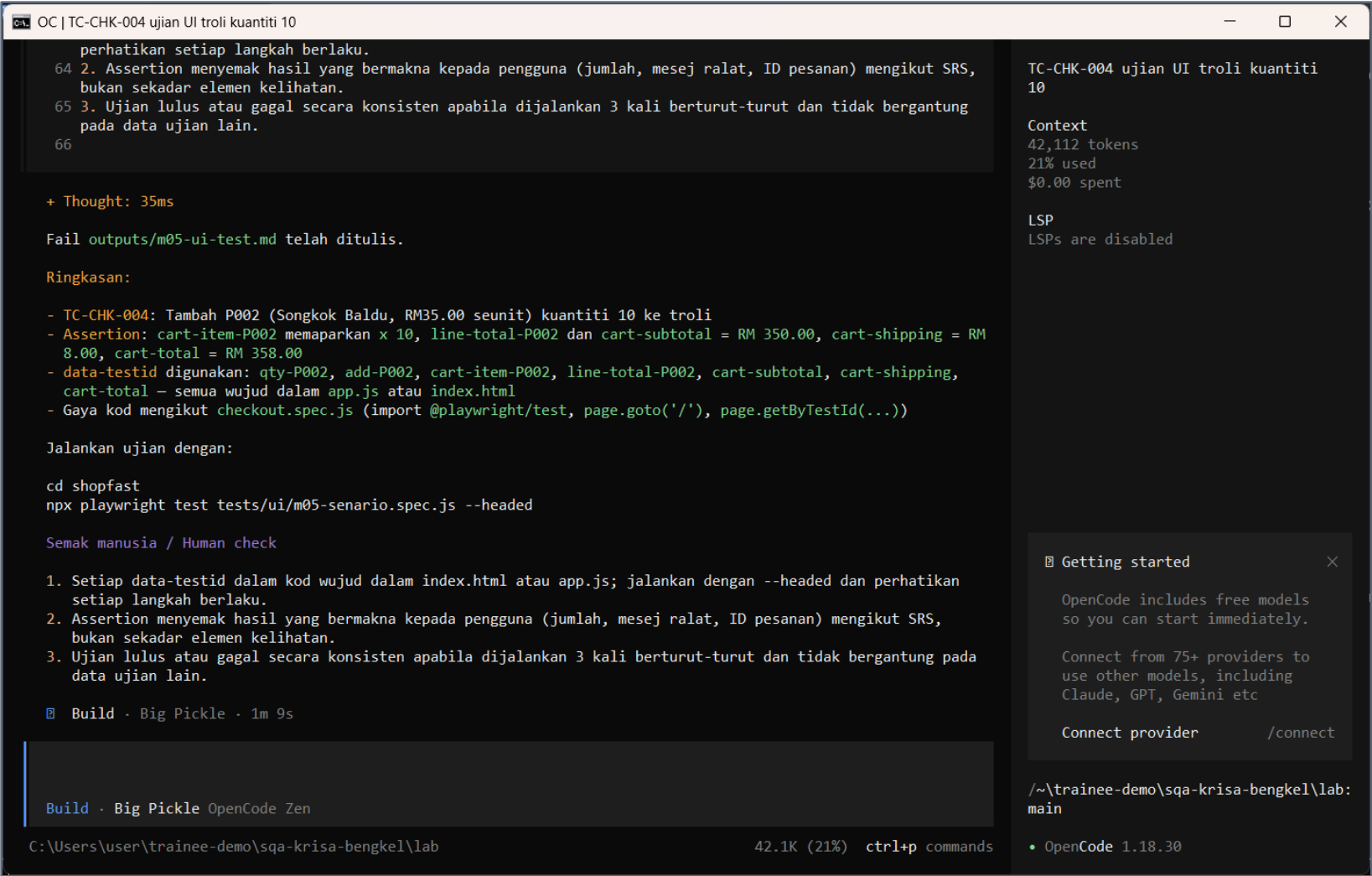
Allow once. The test and instructions are saved in outputs/m05-ui-test.md.

Allow once. Ujian dan arahan disimpan dalam outputs/m05-ui-test.md.

TYPE IN OPENCODE

Salin / Copy

Allow once (Enter)



W05 · STEP 3 OF 5 · LANGKAH 3

Save the code block as a spec file and run it. A failure here is a real finding to reproduce and report in Module 6.

Simpan blok kod sebagai fail spec dan jalankan.
Kegagalan di sini ialah penemuan sebenar untuk dihasilkan semula dan dilaporkan dalam Modul 6.

TYPE IN POWERSHELL

[Salin / Copy](#)

```
$md = Get-Content outputs\m05-ui-test.md -Raw
[regex]::Match($md,
'(?s)```(?:js|javascript)\s*(.*?)```').Groups[1].Value
| Set-Content -Encoding utf8 shopfast\tests\ui\m05-
senario.spec.js
cd shopfast
npx playwright test tests/ui/m05-senario.spec.js --
reporter=list
```

```
SQA-SHOT-23cd5a
Expected substring: "x 10"
Timeout: 5000ms
Error: element(s) not found

Call log:
- Expect "toContainText" getByTestId('cart-item-P002') with timeout 5000ms
- waiting for getByTestId('cart-item-P002')

12 |     await page.getByTestId('add-P002').click();
13 |
> 14 |     await expect(page.getByTestId('cart-item-P002')).toContainText('x 10');
    |                                                    ^
15 |     await expect(page.getByTestId('line-total-P002')).toHaveText('RM 350.00');
16 |     await expect(page.getByTestId('cart-subtotal')).toHaveText('RM 350.00');
17 |     await expect(page.getByTestId('cart-shipping')).toHaveText('RM 8.00');
    at C:\Users\user\trainee-demo\sqa-krisa-bengkel\lab\shopfast\tests\ui\m05-senario.spec.js:14:52

attachment #1: screenshot (image/png)
test-results\m05-senario-TC-CHK-004-tam-295b1--troli-menerima-kuantiti-10-chromium\test-failed-1.png

Error Context: test-results\m05-senario-TC-CHK-004-tam-295b1--troli-menerima-kuantiti-10-chromium\error-context.md

attachment #3: trace (application/zip)
test-results\m05-senario-TC-CHK-004-tam-295b1--troli-menerima-kuantiti-10-chromium\trace.zip
Usage:

    npx playwright show-trace test-results\m05-senario-TC-CHK-004-tam-295b1--troli-menerima-kuantiti-10-chromium\trace.zip

1 failed
[chromium] > tests\ui\m05-senario.spec.js:4:1 > TC-CHK-004 tambah P002 kuantiti 10 ke troli dan semak troli menerima kuantiti 10
PS C:\Users\user\trainee-demo\sqa-krisa-bengkel\lab\shopfast>
```

W05 · STEP 4 OF 5 · LANGKAH 4

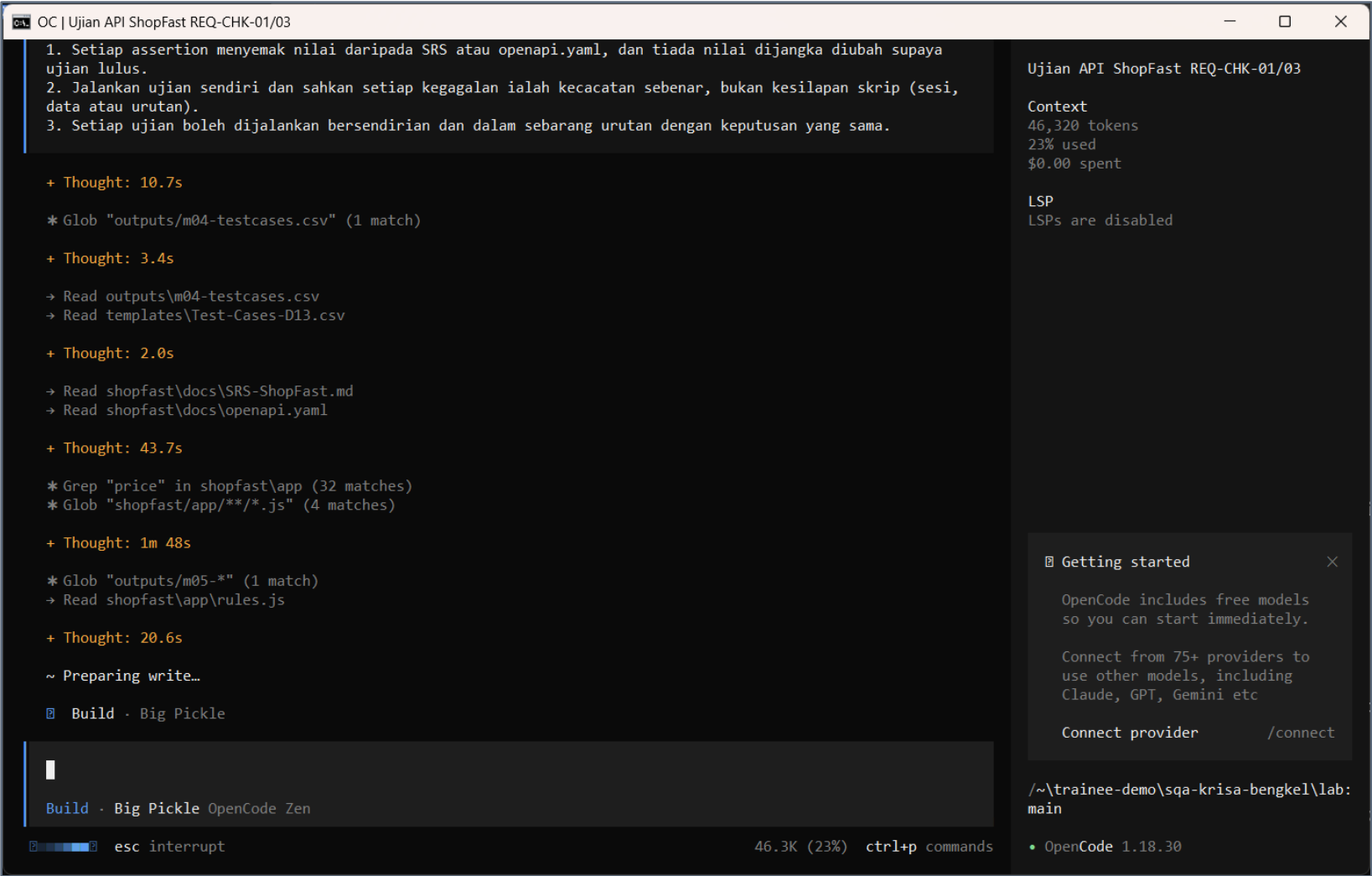
Back in opencode, run /gen-api-tests from the OpenAPI contract. The AI reads the documents first and may pause at Preparing write. When the Permission required dialog appears, read the preview and choose Allow once.

Kembali ke opencode, jalankan /gen-api-tests daripada kontrak OpenAPI. AI membaca dokumen dahulu dan mungkin berhenti seketika pada Preparing write. Apabila dialog Permission required muncul, baca pratonton dan pilih Allow once.

TYPE IN OPENCODE (BUILD)

Salin / Copy

/gen-api-tests REQ-CHK-01 REQ-CHK-03



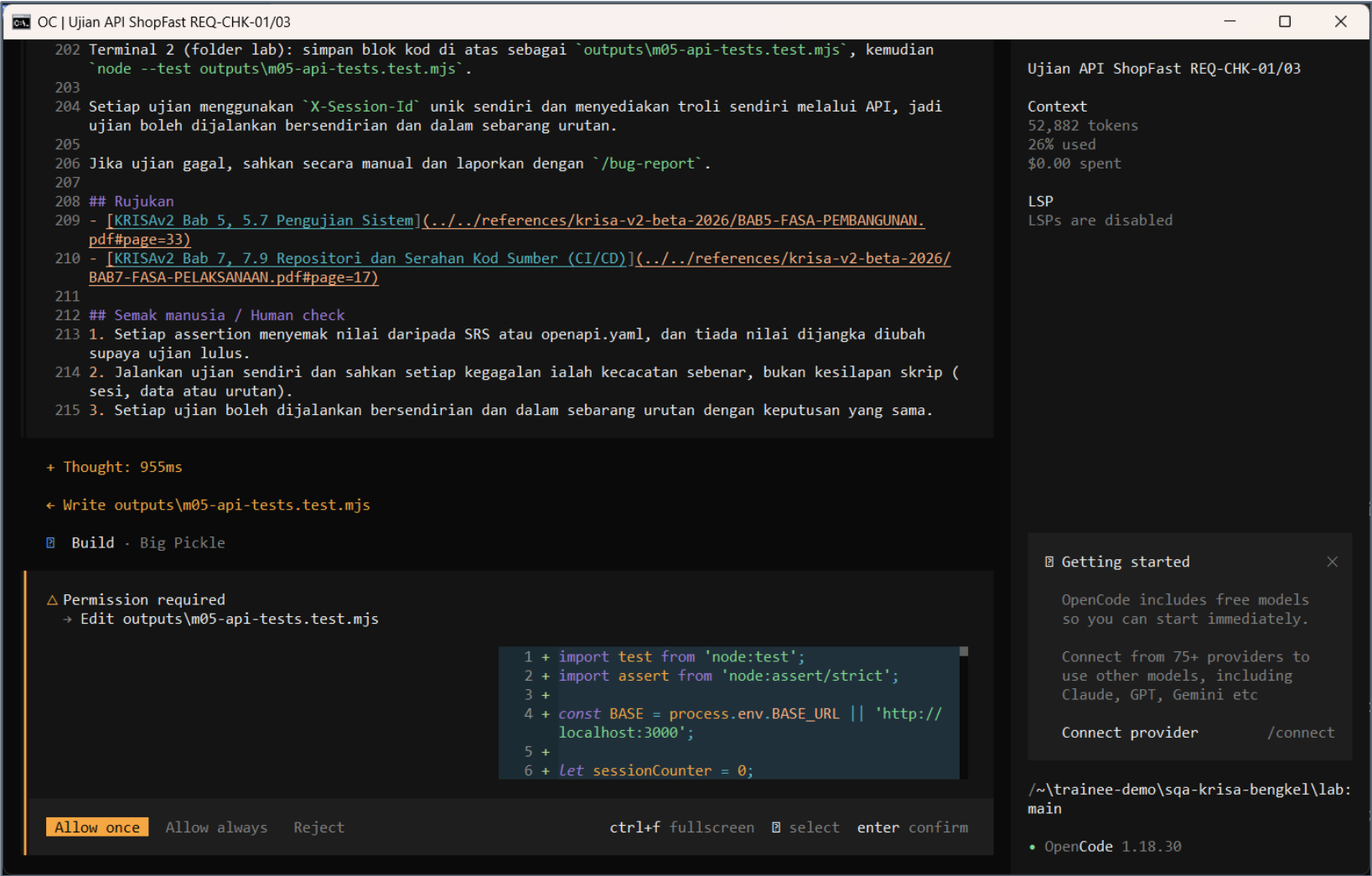
Allow once. Review the generated API tests before using them.

Allow once. Semak ujian API yang dijana sebelum digunakan.

TYPE IN OPENCODE

Salin / Copy

Allow once (Enter)



Human check before you accept the output / Semakan manusia

The test selects elements by data-testid that really exist

Ujian memilih elemen melalui data-testid yang benar-benar wujud

A failing test is a finding: reproduce it by hand before logging a bug

Ujian yang gagal ialah penemuan: hasilkan semula secara manual sebelum log pepijat

Generated tests are reviewed like code before they are committed

Ujian yang dijana disemak seperti kod sebelum di-commit

The AI draft becomes evidence only after this check. Record who checked it and when (Module 7 audit trail).

If something goes wrong / Jika ada masalah

opencode is not recognised

Close and reopen PowerShell so PATH refreshes. Still failing: `powershell -ExecutionPolicy Bypass -File .\setup\install-opencode.ps1 -Portable`

Scripts are disabled on this system

`Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`, or run `opencode.cmd` instead of `opencode`

The model is slow or stops

Wait up to 2 minutes. Press Esc to interrupt, then send the prompt again. Try another free model with `/models`

Permission required dialog

Read the preview. Choose Allow once for files in `outputs/`. Reject anything outside the lab folder

File or command not found

Check the status bar path ends with `\lab`. Commands and `@files` are relative to the lab folder

Answer uses Indonesian words or dashes

Reply: Ikut AGENTS.md peraturan 1 dan 14, tulis semula dalam Bahasa Melayu Malaysia

Network or proxy error

`$env:HTTPS_PROXY = "http://proxy:port"` and `$env:NO_PROXY = "localhost,127.0.0.1"` before starting `opencode`

Port 3000 already in use

`$env:PORT = 3001` before `npm start`, and use `http://localhost:3001`

Abbreviations / Singkatan (1 of 2)

Short forms used in this workshop, in plain words. / Singkatan dalam bengkel ini, dengan bahasa mudah.

Abbreviation / Singkatan	Stands for / Maksud penuh	In plain words / Maksud mudah
AI	Artificial Intelligence	Software that can answer questions and write text or code by learning patterns from large amounts of data. <i>Kecerdasan Buatan. Perisian yang boleh menjawab soalan dan menulis teks atau kod berdasarkan pola daripada data yang banyak.</i>
API	Application Programming Interface	A doorway that lets one program ask another program for data or an action, without using a screen. <i>Pintu yang membolehkan satu program meminta data atau tindakan daripada program lain tanpa melalui skrin.</i>
KRISA	Kejuruteraan Sistem Aplikasi Sektor Awam	The Malaysian public sector guide for engineering application systems. KRISAv2 is the 2026 beta edition. <i>Buku Panduan Kejuruteraan Sistem Aplikasi Sektor Awam. KRISAv2 ialah edisi beta 2026.</i>
MyIPO	Perbadanan Harta Intelek Malaysia	Intellectual Property Corporation of Malaysia, the host of this training venue. <i>Perbadanan Harta Intelek Malaysia, tuan rumah lokasi latihan ini.</i>
PPrISA	Panduan Pengurusan Projek ICT Sektor Awam	The Malaysian public sector guide for managing ICT projects. Codes such as PPrISA05 are its templates. <i>Panduan Pengurusan Projek ICT Sektor Awam. Kod seperti PPrISA05 ialah templatnya.</i>

Abbreviations / Singkatan (2 of 2)

Short forms used in this workshop, in plain words. / Singkatan dalam bengkel ini, dengan bahasa mudah.

Abbreviation / Singkatan	Stands for / Maksud penuh	In plain words / Maksud mudah
REQ	Requirement	Prefix in requirement IDs such as REQ-CHK-01. CHK stands for checkout. <i>Keperluan. Awalan dalam ID keperluan seperti REQ-CHK-01. CHK bermaksud checkout.</i>
SQA	Software Quality Assurance	All the planned work that gives confidence the software meets its quality requirements. <i>Jaminan Kualiti Perisian, iaitu semua kerja terancang yang memberi keyakinan bahawa perisian memenuhi keperluan kualiti.</i>
TC	Test Case	Prefix in test case IDs such as TC-CHK-004. <i>Kes Ujian. Awalan dalam ID kes ujian seperti TC-CHK-004.</i>
TUI	Text User Interface	A full-screen program that runs inside a terminal window, like opencode. <i>Antara muka pengguna berasaskan teks, iaitu program skrin penuh dalam tetingkap terminal seperti opencode.</i>
UI	User Interface	The screens, buttons and forms that people use. <i>Antara muka pengguna, iaitu skrin, butang dan borang yang digunakan orang.</i>

Links / Pautan

[Web guide with copyable prompts](#)

[Module hub \(slides, references, templates\)](#)

[Example outputs from the real run](#)

[Lab README and command list](#)

[AGENTS.md rules used by the assistant](#)

[opencode documentation](#)

[opencode TUI commands and keys](#)

[OpenCode Zen free models and data policy](#)

[Bab 5, 5.7 Pengujian Sistem F4.3 \(p.33\)](#)

[Bab 5, Langkah 5 Laporan Ujian Sistem \(Jadual 5.6\) \(p.39\)](#)

[Bab 5, 5.6.7 Amalan terbaik \(Git, piawaian kod\) \(p.28\)](#)